

Humanoid Sensors and Actuators - Tutorial 3: Part 2

Design and Fabrication of Origami-inspired Soft Actuator

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5 Redesign and Fabrication of a Vacuum Actuator

Based on the design review in T3.1, you will now redesign and fabricate a novel origami-inspired vacuum actuator in your group. You may use the same materials as in T3.1; however, you are encouraged to explore different folding patterns, geometries, and configurations in order to achieve different deformation modes and improved performance.

Before fabricating the final actuator, we recommend that you first test your folding pattern using regular paper. After confirming the basic deformation mechanism, you can fabricate the actuator using stiffer paper or other available materials.

Questions

- (5 points) Describe the deformation that you expect when a vacuum is applied to your new actuator.
- (5 points) Which geometric parameters define the deformation of your actuator?
- (5 points) Describe the design of your new actuator, including its folding pattern, geometric configuration, and expected deformation mode. How does this design differ from the actuator fabricated in T3.1?
- (5 points) Describe the fabrication process of your new actuator. What challenges did you encounter during fabrication, and how did you overcome them?

6 Evaluation of the Actuator

In addition to the deformation measurement performed in T3.1 (see Section 3), you will evaluate the force output of your actuator using a spring scale. Please note that a spring scale measures force along a single axis and extends when a tensile force is applied.

If your actuator produces bending or twisting motion, you may need to design a simple mechanism that converts this motion into a linear displacement or tensile force measurable by the spring scale. For example, in the case of a twisting actuator, you can attach a string to the end of the actuator and connect it to the spring scale. When the actuator twists, it pulls the string and extends the spring scale, allowing you to estimate the force output. If torque is evaluated, clearly explain how the measured force is converted into torque, including the moment arm used in the calculation.

Questions

- (5 points) Plot the relationship between the removed air volume and the deformation of the your original actuator.
- (5 points) Describe the measurement setup used to evaluate the force or torque output. What challenges did you encounter during the measurement, and how did you overcome them?
- (5 points) Plot the relationship between an input parameter and the force or torque output of the actuator. In addition, describe the reason that you chose the input parameter for this analysis.

7 Summary

In this tutorial, you designed and fabricated an origami-inspired vacuum actuator using paper and plastic bags. You then evaluated the deformation and force output of your actuator by applying a vacuum and measuring its displacement and force output. Through this process, you learned about the design and fabrication of soft actuators, as well as the practical challenges involved in their experimental evaluation. You also explored how actuator geometry, folding patterns, material properties, and measurement methods influence actuator performance.

Questions

- (5 points) If you could replace one of the materials used in this tutorial with a commercial product, what would be worth purchasing? Describe the product and explain how it could improve the actuator performance or the evaluation process.
Hint: E-commerce platforms such as RS, Conrad and Amazon
- (5 points) Reflecting on the design, fabrication, and evaluation process, what did you find most challenging? How did you overcome these challenges, and what would you do differently if you repeated this tutorial?